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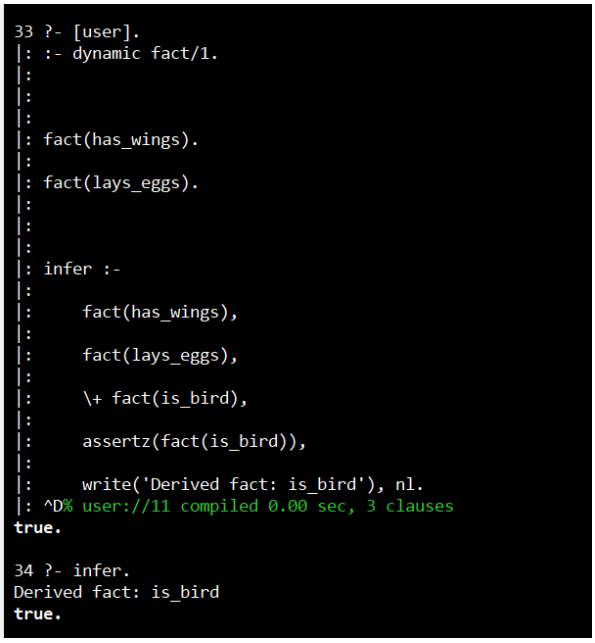
Ex 29: Forward Chaining in Prolog:

PROLOG CODE:

```
% Base facts
fact(has_wings).
fact(lays_eggs).
% Deduction rules
infer :-
  fact(has_wings),
  fact(lays_eggs),
  \+ fact(is_bird),
  assertz(fact(is_bird)),
  write('Derived fact: is_bird'), nl.
```

OUTPUT:



```
33 ?- [user].
|: :- dynamic fact/1.
|:
|:
|:
|: fact(has_wings).
|:
|: fact(lays_eggs).
|:
|:
|:
|: infer :-
|:
|:     fact(has_wings),
|:
|:     fact(lays_eggs),
|:
|:     \+ fact(is_bird),
|:
|:     assertz(fact(is_bird)),
|:
|:     write('Derived fact: is_bird'), nl.
|: ^D% user://11 compiled 0.00 sec, 3 clauses
true.

34 ?- infer.
Derived fact: is_bird
true.
```